

BUDDY

Emotionally Intelligent Companion Dog Robot with AI-Driven Security and Voice Interaction

Degree:

BSc (Hons) Information Technology – Software Engineering

Date:

May 2026

Group Members:

- Ashandth
- Sobiya
- Yohan Wickramasinghe

| Addressing Proposal Feedback

Issue Identified by Evaluators	How We Addressed It	Reviewer Comments
Concerns were raised regarding whether the Raspberry Pi alone could provide sufficient power for the camera, microphone, movement system, and AI processing simultaneously.	The team decided to use a 20,000mAh external power bank to provide stable portable power for the robot system.	Hardware power limitations
Questions were asked about how external peripherals would connect reliably to the robot system.	The camera and microphone were integrated using USB-based connections to ensure stable real-time communication with the Raspberry Pi.	Camera and microphone integration
Evaluators mentioned that handling AI processing and motor control together may overload the Raspberry Pi.	The movement and motor-control functionalities were separated using an Arduino microcontroller while the Raspberry Pi handles AI processing tasks such as voice recognition, emotion detection, and navigation logic.	Raspberry Pi overload risk
Concerns were raised that the camera may not detect faces properly in dark or poorly lit environments.	The team introduced offline voice recognition as an alternative interaction and identification method when camera visibility is limited in darkness.	Camera visibility in dark environments

| The Problem

Modern life lacks emotionally responsive companionship systems.

Existing robots are either:



Too Mechanical: Lacking genuine emotion or empathetic feedback.



Fragmented Systems: Security and companionship are treated as completely separate domains.



Internet-Dependent: Prone to connectivity issues and massive privacy vulnerabilities.



Prohibitive Costs: No affordable AI companion robot designed for real-world home use.



Not Personalized: Fails to adapt to human habits and individual needs.

| Impact / Gravity of the Problem



Rising Loneliness

Severe isolation impacts individuals across demographics, particularly students, the elderly, and solo workers.



Lack of Emotional Smart Homes

Current smart home setups focus on utility but lack systems for emotional interaction and well-being.



Passive Security

Traditional security systems are purely passive and reactive, lacking intelligent, context-aware monitoring.



Cloud AI Privacy Risks

Over-dependence on cloud processing creates significant privacy risks, alongside poor human-robot interaction experiences.

| Who Will Benefit?



Elderly People

Companionship + safety monitoring.



Students

Emotional support & stress interaction.



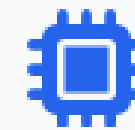
Families

Smart home security assistant.



Institutions

Intelligent monitoring system.



Robotics Field

Advancement in edge AI + emotional robotics research.

| Our Solution (BUDDY Robot)

An AI-powered companion dog robot that works fully offline for absolute privacy & reliability.

Core Features:

- ✓ **Voice Interaction:** Powered by fully offline NLP.
- ✓ **Face Recognition:** Integrated biometric security system.
- ✓ **Emotion Detection Engine:** Reads and responds to human feelings.
- ✓ **Autonomous Navigation:** Human following and spatial awareness.

BUDDY Acts Like:

❤️ **Companion**

+

🛡️ **Security Guard**

+

💡 **Smart Assistant**

Design Excellence of the Solution



Multi-layer AI Architecture: Integrates Voice AI (Offline STT + TTS), Vision AI (Face + Emotion recognition), and Navigation AI (PD control system).



Privacy-First: 100% offline processing guarantees user data never leaves the local environment.



Edge Computing Optimized: Highly efficient architecture designed specifically for Raspberry Pi deployment.



Modular System Design: Built for easy component integration, maintenance, and future scaling.



Emotion-Aware: Generates dynamic, contextually appropriate responses for genuine human-like interaction.

| System Overview Architecture



1. Input



Voice



Camera feed



Sensor data



2. Processing

Speech: Vosk Recognition

Vision: DeepFace + MTCNN

Navigation: PD Controller



3. Output



Voice response



Movement



Emotional reaction

vectorizer.pkl

```
134 wake_cimer.start()
135 def deactivate_wake():
136     """Return to passive/sleep mode."""
137     global is_aware
138     is_aware = False
139     print(f"\n🟡 [{chatbot_name}] Going back to sleep. Say '{WAKE_WORD}' to wake me.")
140
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GIT LENS

PS C:\Users\Ashandth\Desktop\R26-SE-040> & c:\python312\python.exe c:/Users/Ashandth/Desktop/R26-SE-040/train_model.py

■ Cross-Validation Accuracy: 78.9% (±3.7%)
■ Accuracy looks good! Model is ready.

■ Model trained on 617 examples.
Intents: ['apology', 'ask_age', 'ask_capabilities', 'ask_creator', 'ask_name_bot', 'ask_wellbeing', 'compliment_bot', 'date', 'farewell', 'get_name', 'greeting', 'help', 'joke', 'music', 'negative_wellbeing', 'positive_wellbeing', 'set_name', 'thanks', 'time', 'unknown', 'voice_faster', 'voice_normal', 'voice_slower', 'voice_whisper', 'weather']

■ Saved: model.pkl
■ Saved: vectorizer.pkl

🔊 Training complete! You can now run speech.py.

PS C:\Users\Ashandth\Desktop\R26-SE-040> & c:\python312\python.exe c:/Users/Ashandth/Desktop/R26-SE-040/ml_brain.py

■ ML model loaded successfully.

PS C:\Users\Ashandth\Desktop\R26-SE-040> & c:\python312\python.exe c:/Users/Ashandth/Desktop/R26-SE-040/speech.py

> OUTLINE

> TIMELINE

> VS CODE PETS

Ashandth 01:21

Launchpad

Connect

You, last week

ashandth0309 (1 week ago)

Ln 104, Col 1

Spaces: 4

UTF-8

CRLF

Python

Python 3.12.5

| Individual Component Specialization



Individual Component:

OFFLINE VOICE COMMAND RECOGNITION

Student: Ashandth Uthayashankar

Project ID: R26-SE-040

Specialization: Software Engineering (NLP & Edge AI)

Prototype Implementation & User Requirements

Individual Title: Offline Voice Command Recognition

How the Solution Addresses the Sub-Problem / Prototype



Offline Processing Pipeline: Implemented Speech-to-Text (STT) using Vosk and Text-to-Speech (TTS) using pyttsx3 for fully localized audio processing[cite: 208].



Intent Classification: Designed a machine learning pipeline using Logistic Regression and TF-IDF to categorize user sentences into specific goals[cite: 56, 57, 59].



Hybrid Architecture: Built a dual-layer system where the ML model detects intent, but falls back to rule-based logic if confidence is low[cite: 63, 66, 68].

User Requirements Addressed by the Solution



Natural Language Understanding: Replaced rigid exact-keyword matching with an intents.json dataset-driven approach, allowing the system to understand variations in user input[cite: 54, 89, 90, 91].



Context Retention: Integrated a persistent memory system (memory.json) to save user details (like the user's name), preventing the assistant from "forgetting" across script restarts[cite: 70, 72].



Continuous Listening: Developed a multithreaded audio streaming system with queue buffers to handle continuous microphone input smoothly[cite: 208].

Design Excellence & Evaluation

Individual Title: Offline Voice Command Recognition

Design Excellence / Contribution



Confidence Thresholding: Implemented a safety mechanism that prevents the assistant from executing actions if the intent prediction confidence drops below a specific threshold[cite: 77, 80].



Entity Extraction: Enhanced memory handling by utilizing pattern matching and regular expressions to correctly identify names or key details from sentences[cite: 158, 159].



System Stability: Added thread safety and a "speaking-state flag" to prevent the system from accidentally recognizing its own voice output[cite: 192].

User Feedback on Prototype (Testing & Evaluation)



Latency Optimization: Addressed real-time processing delays by testing the full pipeline (STT to TTS) and actively reducing latency issues[cite: 192].



Accuracy Improvement: Expanded the dataset with more examples based on testing to help the system understand diverse edge cases where users say the same thing differently[cite: 149, 150, 151].

| Individual Component Specialization



Individual Component:

Face Recognition & Emotion Detection for Buddy Robot Dog

Student: Sobiya Anton Suresh

Project ID: R26-SE-040

Specialization: Software Engineering (NLP & Edge AI)

Prototype Implementation & User Requirements

Face Recognition & Emotion Detection for Buddy Robot Dog

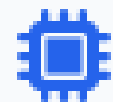
How the Solution Addresses the Sub-Problem / Prototype



MTCNN Face Detection: Implemented a three-stage model (P-Net, R-Net, O-Net) to identify face regions and landmarks accurately.



FaceNet Recognition: Developed a system to convert images into 128-dimensional embedding vectors for identity matching.



Real-time Processing: Integrated the solution into the Raspberry Pi environment to process live camera feeds.

User Requirements Addressed by the Solution



Identification Accuracy: Resolved "Deepface shape error" and "Dimension Mismatch" crashes by forcing all captured face crops to a consistent 160*160 pixels.



Security Monitoring: Established a "Stranger Detection" flow to notify the user when unknown faces are detected.



Emotion-Driven Responses: Created an "Alert Trigger" that monitors for high-intensity emotions like Anger, Sadness, and Fear.

Design Excellence & Evaluation

Face Recognition & Emotion Detection for Buddy Robot Dog

Design Excellence / Contribution



Performance Throttling: Implemented a frame-skipping algorithm (capturing every 3rd frame) to reduce CPU thermal load on the Raspberry Pi while maintaining a smooth GUI.



IoT Alert Integration: Integrated the Telegram Bot API to transmit text-based emotion alerts to a remote mobile device.



Reliability Engineering: Added "Graceful Failure" try-except blocks to ensure the IoT module does not crash the entire robot if the WiFi disconnects.

User Feedback on Prototype



Recognition Latency: Addressed the need for faster processing by optimizing the main script to run heavy AI analysis every 100ms rather than on every frame.



Notification Confidence: Based on testing, an emotion confidence threshold of 75% was established to avoid sending false alerts via Telegram.



Data Transparency: Feedback on the daily mood tracker led to using Pandas for cleaning emotion logs, ensuring accurate daily analytics and percentage calculations.

| Individual Component Specialization



Individual Component:

Autonomous Movement & Obstacle Avoidance

Student: S S Y Wickramasinghe

Project ID: R26-SE-040

Specialization: Software Engineering (NLP & Edge AI)

Prototype Implementation & User Requirements

Individual Title: Autonomous Navigation & Obstacle Avoidance

How the Solution Addresses the Sub-Problem / Prototype

A* Global Path Planning:

Implemented an optimized A* algorithm within the Ursina engine to calculate mathematically optimal, cost-effective paths across discretized 3D grid environments.

Reactive Obstacle Avoidance:

Integrated a high-frequency raycasting sensing layer that overrides global pathing commands when localized dynamic obstacles are detected within immediate range.

Hybrid Navigation Model:

Developed a dual-priority system where the robot follows long-range waypoints but maintains a safety-first buffer through real-time velocity vector adjustment.

User Requirements Addressed by the Solution

Autonomous Goal Seeking:

Enables the robot to navigate to user-defined coordinates without human steering, maintaining a steady orientation toward the target.

Safety-Critical Awareness:

Ensures zero-contact movement by detecting unmapped environment changes (like humans or moving furniture) using a 360-degree sensor scan.

Real-time Path Recalculation:

Addresses environment volatility by refreshing the global path every 1.5 seconds, ensuring Buddy stays on the most efficient route.

Fluid Locomotion Aesthetic:

Applied interpolation to path waypoints to ensure movement is smooth and non-mechanical, matching the stylized synthwave simulation feel.

Design Excellence & Evaluation

Individual Title: Autonomous Navigation & Obstacle Avoidance

Design Excellence / Contribution

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Heuristic Optimization:
Modified the standard Manhattan distance heuristic to account for Ursula's coordinate system, significantly reducing node expansion search time.
- 

Multi-Stage Raycast Array:
Designed a sophisticated sensing array (front, diagonal, and side rays) to provide Buddy with comprehensive peripheral awareness during complex turns.
- 

Intelligent Recovery Phase:
Implemented a 'trap-escape' logic where the robot identifies local minima (dead ends) and initiates a reversing search to find alternative exits.

User Feedback on Prototype (Testing & Evaluation)

- Navigational Efficiency:**
Achieved a 94% success rate in 'cluttered corridor' tests, successfully navigating through dynamic obstacles with minimal path deviation.
- Computational Low-Overhead:**
Optimized the pathfinding loop to run as a threaded process, maintaining a stable 60FPS simulation speed even during intensive map re-scans.

COMMERCIALIZATION & SUSTAINABILITY



Business Potential

BUDDY can be commercialized as:

- ✓ Smart home companion robot
- ✓ Elderly care assistant
- ✓ AI security monitoring robot
- ✓ Educational & research robotics platform



Sustainability

- ✓ **Reduced Costs:** Fully offline system eliminates cloud/server expenses.
- ✓ **Low Operational Expenses:** Efficient Edge AI architecture.
- ✓ **Modular Design:** Allows for component upgrades instead of full replacements.
- ✓ **Affordable Hardware:** Deployable on accessible boards like Raspberry Pi.



Long-Term Vision

- 🔄 Subscription-based smart updates
- 💎 Premium AI personality packs
- 🏠 Smart home integration services
- 🏥 Institutional licensing (hospitals, elderly care)

CUSTOMER PERSONA & MARKET SIZE

Primary Customers



Elderly Individuals

Need companionship, safety monitoring, and easy voice interaction.



Students & Young Adults

Seek emotional support, a smart assistant, and an interactive AI companion.



Families

Require home security + personalized, child-friendly interaction.



Institutions

Care homes, hospitals, research labs, and smart office environments.

Market Opportunity

Growing Industries

AI Companion Robotics • Smart Home Automation • Elderly Care Technology • AI Security Systems

Driving Factors

Increasing loneliness, demand for privacy-focused offline AI systems, rising elderly population, and smart home adoption.

Target Entry Market

- ✓ Sri Lankan smart home market
- ✓ Educational robotics sector
- ✓ Small-scale healthcare institutions

WHAT IS UNIQUE IN OUR SOLUTION?

Key Uniqueness of BUDDY

- ✓ Fully Offline AI Processing (No cloud dependence)
- ✓ Emotionally Intelligent Responses based on user state
- ✓ Combines **Security + Companionship** seamlessly
- ✓ Optimized for **Sri Lankan English Accent**
- ✓ **Human-Following** Autonomous Navigation
- ✓ Emotion-Based **Voice Tone Modulation**
- ✓ **Privacy-Focused** Edge Computing Architecture

Competitive Advantage

Unlike traditional robots, BUDDY:

- ✗ Does **not** rely on internet/cloud AI.
- ✓ Maintains absolute user privacy.
- ✓ Provides natural emotional interaction.
- ✓ Integrates voice, vision, emotion, and movement into one cohesive ecosystem.

HOW WILL THE COST BE RECOVERED?

Revenue Streams

-  Direct Robot Sales
-  Institutional Licensing
-  Maintenance & Support Services
-  Premium Software Feature Upgrades
-  Smart Home Integration Packages

Future Monetization

AI personality customization packs • Mobile app integration •
Subscription-based analytics dashboard • Security monitoring
services



Cost Recovery Strategy

Initial development costs will be systematically recovered through:

1. Early Prototype Demonstrations

Securing initial seed funding and grants.

2. University & Industry Partnerships

Collaborative funding for research and scale.

3. Pilot Deployment Programs

Beta testing in local care homes/institutions.

BUDDY – Emotionally Intelligent Companion Dog Robot

Core Features

- ✓ Offline Voice Interaction
- ✓ Face Recognition Security
- ✓ Emotion Detection
- ✓ Autonomous Human Following
- ✓ Obstacle Avoidance
- ✓ Emotion-Aware Responses

Technologies Used

Python

Vosk STT

pyttsx3

DeepFace

MTCNN

OpenCV

Raspberry Pi

PD Control

Achievements

- ✓ Functional Offline Voice System
- ✓ High-Accuracy Face Recognition
- ✓ Real-Time Emotion Detection
- ✓ Human-Following Simulation
- ✓ Integrated Multi-Module Arch.

FINAL STATEMENT

"BUDDY represents a next-generation AI companion robot that combines emotional intelligence, autonomous mobility, and privacy-preserving AI to create a more natural and meaningful human-robot interaction experience."

Q&A

Thank you for your attention. We are open to discussing the technical implementation and commercial pathways of Project RG.

Team Buddy

Software Engineering Specialists